PDF INGESTION - CHANGE SUMMARY

What changed

? Added a shared element index for headings, paragraphs, lists, captions, tables, and images. The index points back to existing extracted content instead of duplicating it.
? Added zero-based PDF page indexes and consistent source provenance. Printed page numbers remain null when they cannot be reliably identified.
? Split image bytes from image placements: unique image assets are saved once, while each page occurrence has its own ID, page, and bounding box.
? Made relationship certainty explicit: confirmed, candidate, or unresolved. Nearby text and geometry-based caption links stay candidates, with evidence recorded.
? Added document checks for missing provenance, invalid boxes/references, missing image assets or output files, row/header alignment, and unsupported relationships. Manifest entries now include warning and validation details.
? Retained suspicious table detections and raw rows, but excluded probable page-wide false detections from table-row RAG units. Reused the PDF hash to avoid hashing the same source twice in the coordinated run.

What was already working and kept

The existing pipeline already discovered PDFs, extracted text and images with PyMuPDF, detected tables, inferred some merged and multi-level headers, retained raw rows, flagged table uncertainty, detected possible text/table duplicates, built relationships and retrieval units, and continued after a per-PDF failure. Those functions were extended rather than replaced.

Checks that actually ran

? Unit tests: 23 passed. Command: .\.venv\Scripts\python.exe -m unittest discover -s tests -p "test_*.py"
? Full local ingestion: 7 PDFs processed, 0 failed.
? Output verification: all document JSON, extraction JSON, table JSON, and log files existed; raw table rows matched between table and combined JSON; RAG units resolved to elements; relationships had valid element endpoints and evidence.
? Pylance diagnostics: no diagnostics in the checked changed Python files. Syntax checks passed.
? git diff --check: passed.

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What the real-PDF run found

The seven local manuals produced 35 table detections. Twenty-one were marked as probable page-wide false positives. These were retained in table output for inspection and excluded from table-row retrieval units. The batch recorded 90 validation warnings across successful PDFs; six documents were marked needs_review. These flags mean ?inspect this extraction,? not that the PDF processing failed.

Example retrieval record

From the Operation manual, an extracted table row was represented with its original table element ID and page index:

Text: Main Control
NUMBER: 1; TYPE: Screen; NAME: Main screen

Headers: NUMBER, TYPE, NAME
Row: NUMBER=1; TYPE=Screen; NAME=Main screen

An image retrieval unit also points to its image occurrence and saved asset. If associated text was only nearby on the page, its context status is candidate rather than a confirmed description.

Files changed

Code: src/ingestion/main.py, pdf_extractor.py, pipeline.py, and table_extractor.py. Tests: tests/test_ingestion_pipeline.py. Documentation: README.md and docs/INGESTION_FLOW.md. The generated data/processed outputs were refreshed by rerunning the pipeline.

Still not implemented

Scanned-page OCR and coordinate-based table reconstruction remain future work. Printed page numbering is not inferred. Image/table relationships based on layout remain uncertain candidates. No embeddings, vector database, cloud service, or LLM were added. Review needs_review tables before using them as maintenance guidance.